Blood Gas Homeostasis: Regulation of Heart Rate and Blood Pressure



Function of Circulatory System in Gas Homeostasis

- Heart pumps blood, vessels carry blood
- O_2, CO_2 transported in blood stream
- Heart rate and blood pressure affect how quickly:
  - O_2 is carried from lungs to tissues
  - CO_2 is carried from cells to lungs for removal

Vocabulary:

Heart Rate: Number of times heart beats per minute

Stroke Volume: Amount of blood (mL) leaving heart with each beat

Cardiac Output: Amount of blood leaving the heart each minute

= Heart Rate x Stroke Volume

$$\textbf{Cardiac output}\ (\text{mL/min}) = \textbf{heart rate}\ (\text{beats/min}) \times \textbf{stroke volume}\ (\text{mL})$$

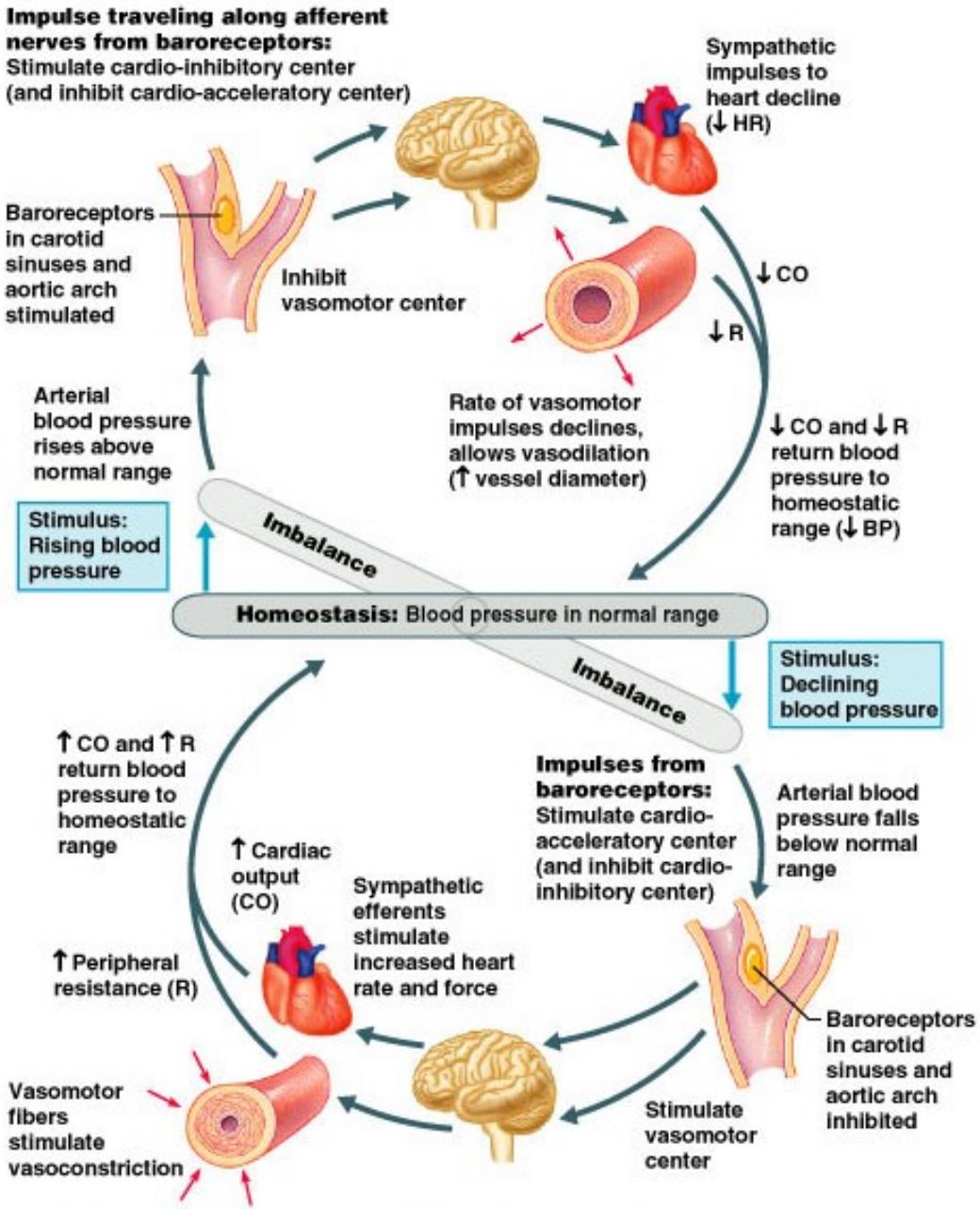
Blood Pressure: force of blood on vessel walls. Depends on:

- Whether heart is contracting (systole) or relaxing (diastole)
- Cardiac Output (as CO increases, BP increases)
- Diameter of blood vessels (constriction increases BP)

Regulation of Heart Rate and Blood Pressure

- Changes in body are detected by receptors eg:
  - Chemical Composition of Blood: Chemoreceptors in Aortic and Carotid Bodies
  - Conscious awareness of changes eg danger/fear
  - Changes in blood volume and pressure: receptors in vessel walls
- Information sent to:
  - Medulla oblongata – cardiac centre (cardiovascular regulating centre)

- Signals sent via Autonomic Nervous System to heart and blood vessels, to affect blood pressure, heart rate, stroke volume, cardiac output



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Homeostasis when Blood Pressure Rises

Stimulus

Rising Blood Pressure

Receptor

Baroreceptors in carotid and aortic arch detect rising BP

Message

Nerve signals

Control Centre / Modulator

Cardiac Centre in Medulla Oblongata

Message

Autonomic NS signalling

Effector

Heart Muscle

Blood Vessels

Response

Decreased Heart Rate and strength of contraction—decreases blood pressure

Vasodilation decreases blood pressure

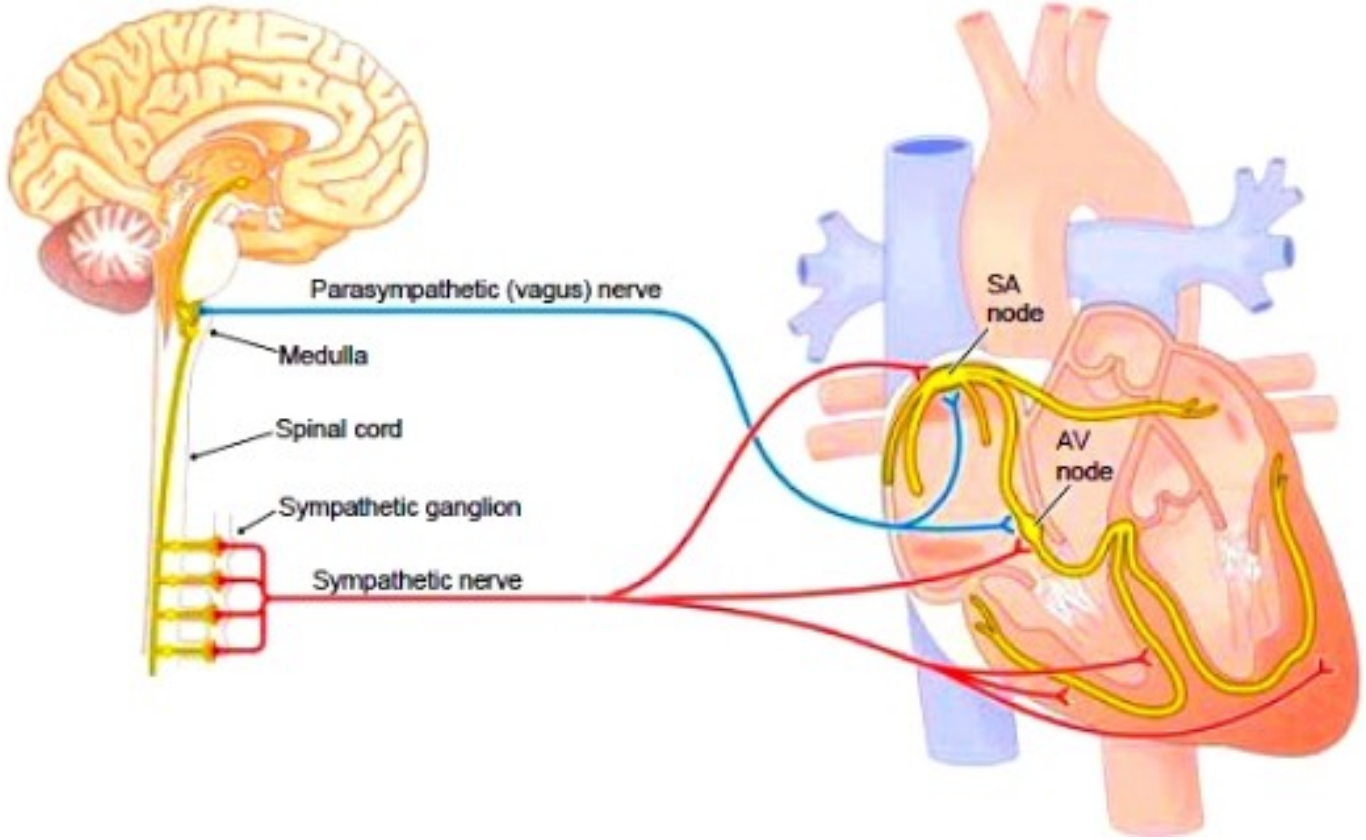
Feedback

Negative Feedback: blood pressure drops to normal, stimulus ceases, process stops



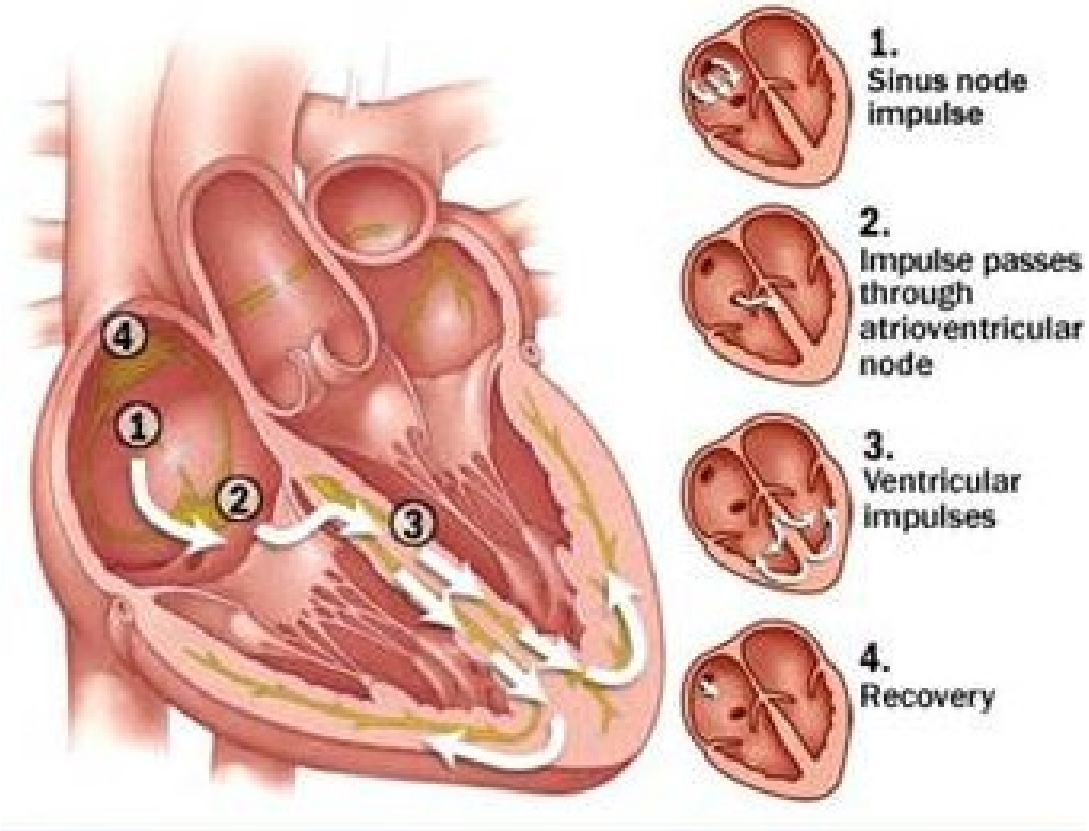
Heart Rate Regulation

- Primarily via autonomic NS stimulation



Coordination of heart beat

- Autonomic NS stimulates receptor within the heart:
  - Sinoatrial Node: “pacemaker” – controls the electrical conduction system that controls heart contraction.



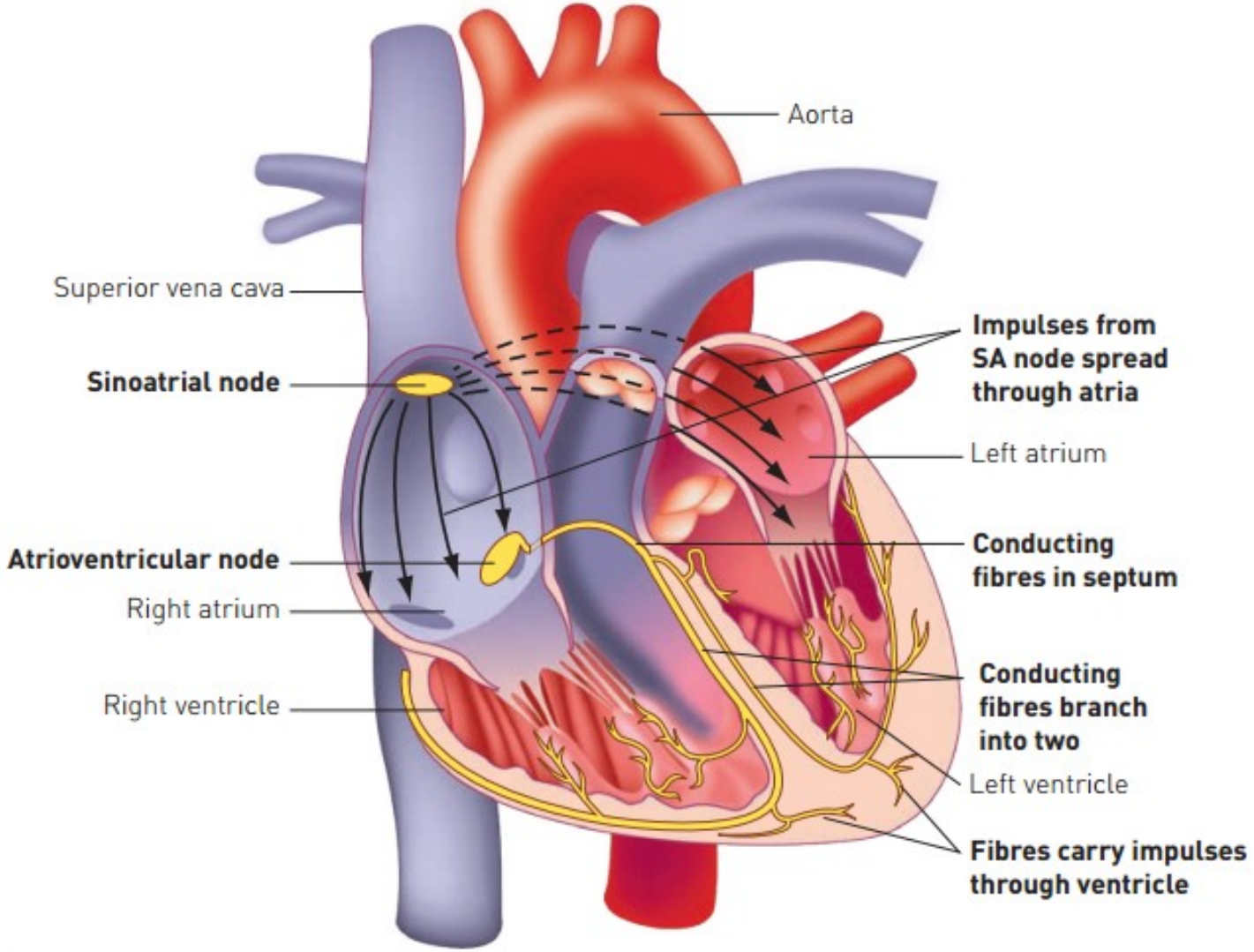


Figure 8.16 The conduction of nerve impulses through the heart

Changes to Blood Flow During Exercise

- Activity of muscles cells uses oxygen, releases carbon dioxide waste
- In order to supply cells with more oxygen, and carry away carbon dioxide, the Cardiac Output increases (along with rate of breathing)
- Blood Vessels to muscles dilate, for better supply
- Vessels in other internal organs constrict, to direct blood to muscles
- Sympathetic stimulation assists with this via release of adrenaline.

Effect of Behaviour

- Conscious input can affect how the body responds
  - Heart Rate
  - Cardiac Output
  - Breathing Rate

- Eg: before an exam or test:
  - Feeling nervous about the test stimulates Sympathetic NS, and Cortisol release:
    - Increase in Blood Glucose to supply muscles
    - Dilation of Vessels to muscles
    - Increase in Respiratory Rate.
  - Even though the test may not involve physical exertion, the body anticipates the “flight or fight” response.